

End-to-End Data Science Report

From data preparation and exploratory analysis to predictive modelling and retention strategy

1,000,000 customer records	31 cleaned variables	9.92% overall churn rate
5 classification models	0.685 best ROC-AUC	63.9% XGBoost churn recall

Primary result

Contract structure, customer satisfaction, complaint history, service-call frequency and late payments were the most consistent churn indicators across both exploratory analysis and predictive modelling.

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Portfolio report based on a cleaned synthetic telecommunications customer dataset.

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How to read this report

The three project notebooks contain the complete code, checks, distributions and intermediate outputs. This report consolidates the most decision-relevant evidence into one recruiter- and stakeholder-facing narrative.

Project deliverables

- Data preparation notebook: cleaning, missing-value treatment, validation and export of the modelling dataset.
- Exploratory analysis notebook and EDA report: business questions, visual evidence and retention hypotheses.
- Feature engineering and modelling notebook: five classifiers, class-imbalance handling, model comparison and interpretation.

[Project repository: Customer-Churn-Analysis-and-Prediction](#)

Executive Summary

This project developed an end-to-end customer churn analysis using one million synthetic telecommunications customer records. The workflow covered data preparation, exploratory analysis, feature engineering, classification modelling, model interpretation and business recommendations. The objective was not only to predict churn, but to determine which customer signals are sufficiently strong, observable and actionable to support retention decisions.

Data quality was strong after documented preprocessing. Five variables contained between 1.99% and 5.00% missingness; numerical values were imputed using medians, with average monthly usage imputed within internet-service groups. No duplicate records were found, all customer identifiers were unique and the final cleaned dataset contained 1,000,000 rows and 31 variables. The outcome was imbalanced, with 9.92% of customers classified as churned.

Exploratory analysis showed that behavioural and account variables carried substantially more standalone signal than demographics. Month-to-month customers recorded a churn rate of 26.5%, compared with 12.7% for one-year and 5.7% for two-year contracts. Churn also rose with repeated complaints, service calls, low satisfaction and late payments. Technical support, online security and broader service adoption were associated with stronger retention, while age, gender, education, marital status, income, payment method, credit score and usage measures showed little practical separation.

Five models were compared: Logistic Regression, Decision Tree, Random Forest, Gradient Boosting and XGBoost. Results were tightly clustered, with ROC-AUC values between 0.6704 and 0.6850. XGBoost achieved the highest ROC-AUC at 0.6850, while Gradient Boosting produced the highest recall at 65.05%. Logistic Regression was effectively tied with XGBoost, demonstrating that model complexity did not materially improve discrimination with the available feature set.

Final analytical conclusion

The model is most useful as a churn-risk ranking tool rather than a deployment-ready binary classifier at the default threshold. Its 63.9% recall captures most churners, but 15.9% precision creates many false positives. The next technical priority is threshold selection using retention costs, customer value and precision-recall analysis.

Retention priorities

- Prioritise month-to-month customers for targeted loyalty and longer-term contract offers.
- Build a service-friction alert using satisfaction, complaints and service-call frequency.
- Test technical-support and online-security bundles among comparable at-risk customers.
- Use repeated late payments as an early warning signal alongside service-experience indicators.
- Avoid demographic-first targeting; focus retention resources on observable behaviour and account structure.

1. Business Context and Project Objectives

Customer churn reduces recurring revenue and can increase the cost of replacing lost customers. A useful churn project therefore needs to move beyond describing who leaves. It should identify warning signals that can be measured early, tested through predictive modelling and translated into realistic retention actions.

The analysis was designed as an individual portfolio project demonstrating the full data-science lifecycle. It combines technical implementation with business interpretation and deliberately separates descriptive association, predictive importance and causal decision-making.

Project objectives

- Clean and validate a large customer dataset while documenting reproducible preprocessing decisions.
- Identify customer segments and behaviours with meaningful differences in churn.
- Engineer a value-related feature and prepare mixed numeric and categorical data for modelling.
- Train and compare linear, tree-based and boosted classification models under class imbalance.
- Interpret the strongest model and assess whether exploratory findings remain important jointly.
- Produce practical retention recommendations while clearly stating limitations and next steps.



Figure 1. End-to-end project workflow. Source: author analysis.

Scope and interpretation

The dataset is synthetic. Results identify associations and predictive patterns within this dataset; they do not prove that changing a contract, service or customer experience variable will causally prevent churn. Customer-facing interventions require controlled testing.

2. Data Preparation and Quality Assurance

The raw dataset contained 1,000,000 customer records and 32 variables. Preparation converted the sign-up date to a datetime type, assessed missingness, imputed incomplete numerical values, checked duplicates and identifier uniqueness, reviewed numerical ranges and category labels, removed the non-predictive customer identifier and exported a cleaned dataset for analysis.

Missing-value treatment

Only five variables contained missing data. Because missingness was limited to 1.99%-5.00%, deleting complete customer records would have removed substantial usable information. Median imputation was therefore applied because it is less sensitive to extreme values than the mean. Average monthly data usage was imputed within internet-service status groups, reflecting the structural difference between customers with and without internet service.

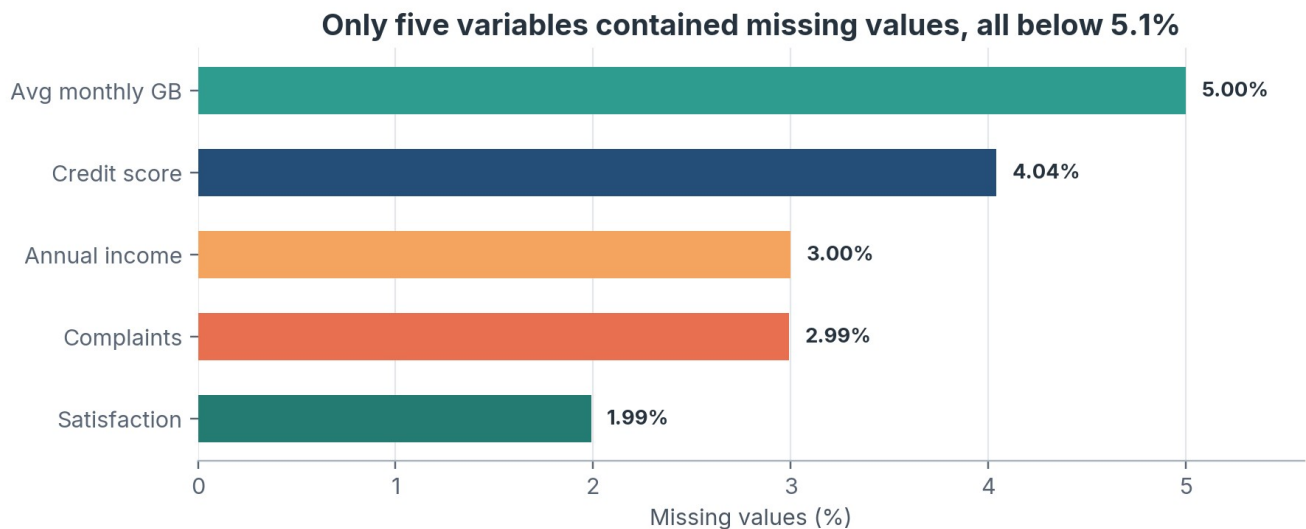


Figure 2. Missingness in the raw dataset. All other variables were complete.

Validation results

Check	Result	Interpretation
Duplicate rows	0	No duplicated customer records detected
Unique customer IDs	1,000,000	Each row represented a distinct customer
Age range	18-90	No obvious invalid ages
Annual income	\$20,000-\$250,000	Within the dataset's defined range
Credit score	300-850	Consistent with conventional scoring bounds
Churn distribution	90.08% retained / 9.92% churned	Class imbalance required non-accuracy metrics
Final dataset	1,000,000 rows x 31 variables	Customer ID removed and cleaned data exported

Data-quality outcome

The cleaned dataset retained all one million customers, removed the identifier not needed for prediction and contained no remaining missing values. Potentially extreme monthly-charge values were retained for investigation rather than removed automatically.

3. Exploratory Analysis

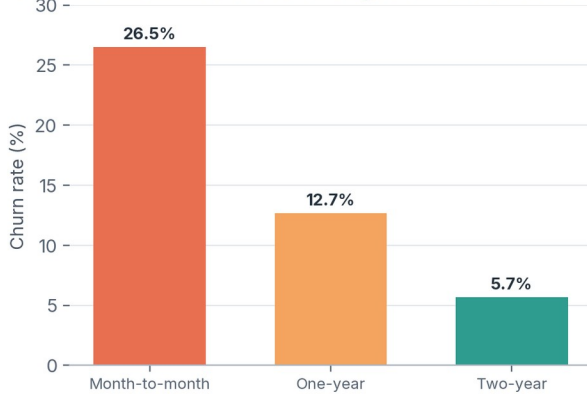
Business Question -> Data Analysis -> Visualisation -> Key Finding

3.1 Contract structure and customer lifecycle

Contract type produced the largest account-level difference in churn. Month-to-month customers churned at 26.51%, more than twice the rate for one-year customers and almost five times the rate for two-year customers. This makes contract structure the clearest high-level segmentation variable, although the relationship may partly reflect customer commitment, tenure, service mix and selection effects.

Tenure showed a smaller lifecycle pattern. Churn was highest during the first six months at 10.39% and generally declined toward approximately 9.2% after four years. Churned customers had a slightly shorter mean tenure than retained customers, but the absolute difference was modest, so tenure alone is not a strong screening variable.

Contract structure is the strongest account-level signal



Churn declines modestly across the customer lifecycle

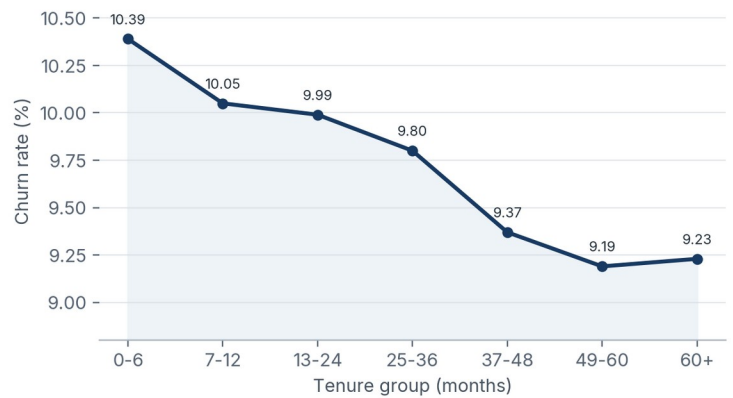


Figure 3. Churn by contract type and tenure group. Contract differences are substantially larger than lifecycle differences.

Business interpretation

Month-to-month customers are the highest-priority segment for retention analysis. New-customer onboarding may provide a secondary opportunity, but tenure should be combined with stronger behavioural and account signals.

3.2 Customer satisfaction and service friction

Service experience produced the clearest actionable behavioural signals. Churn declined from 16.72% at the lowest satisfaction score to 7.48% at the highest, with a noticeable reduction between scores three and four. Complaint history showed a consistent escalation from 8.10% with no complaints to 22.14% with four or more complaints.

Service-call frequency displayed a similar pattern. Churn rose from 7.71% for customers with no calls to 15.03% at five calls and 19.68% at six calls, reaching 31.19% at ten calls. The apparent reversal at eleven and twelve calls is not reliable because those groups contained only 68 and 36 customers.

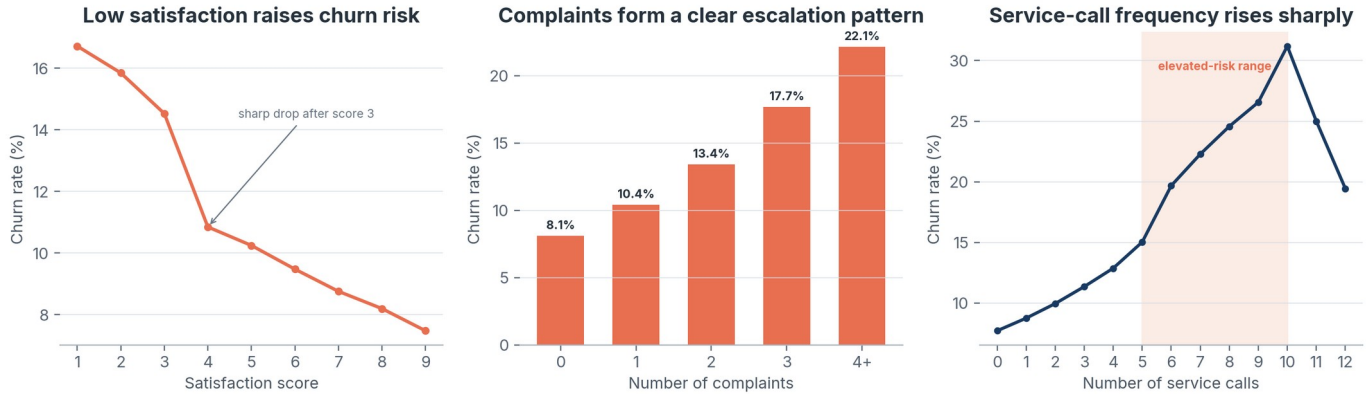


Figure 4. Satisfaction, complaint and service-call patterns. High-count call categories require cautious interpretation.

Potential operational trigger

A combined service-friction alert could flag customers with satisfaction scores of three or below, repeated complaints or approximately five or more service calls. The final rule should be selected using validation data and the cost of false alarms rather than EDA alone.

3.3 Customer value, service adoption and payment behaviour

Direct engagement and raw pricing measures were weak. Churned and retained customers had the same median days since last interaction, nearly identical average usage and only small differences in monthly charges. Total charges were lower among churned customers, which was consistent with their slightly shorter tenure rather than clear evidence of an independent pricing effect.

The engineered monthly charge per service measure produced a clearer gradient. Churn increased from 8.61% in the lowest decile to 11.27% in the highest. Because this feature is mathematically related to monthly charges and service count, it should be interpreted as a value proxy and evaluated alongside its components.

Service adoption was associated with retention: churn fell from 11.04% among customers with one service to 7.37% among customers with six. Technical support and online security showed the strongest individual service differences, while entertainment add-ons showed little separation. Late payments were the clearest financial warning signal, with churn increasing from 9.06% with no late payments to 17.16% with three or more.

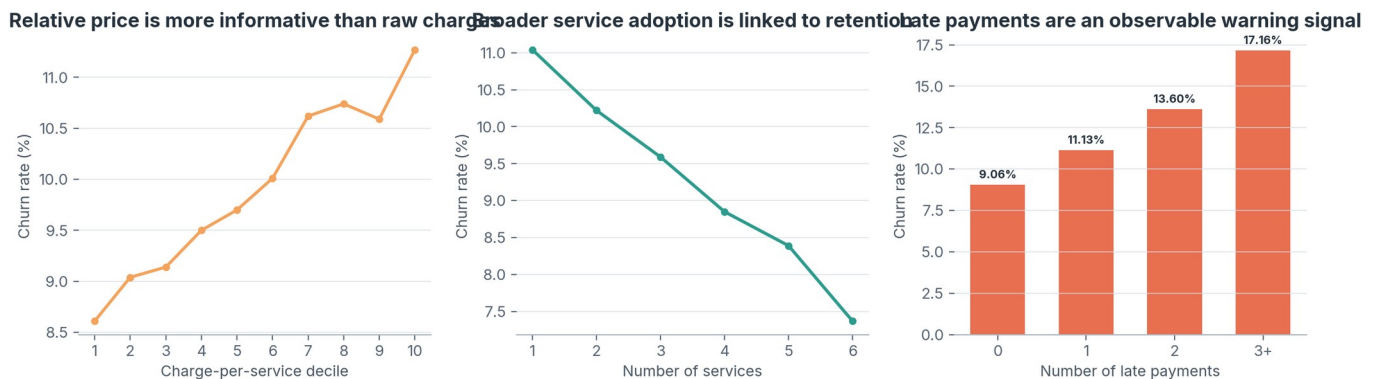


Figure 5. Churn patterns across relative price, service adoption and late payments.

Interpretation

Support-oriented services and broader adoption may improve customer stickiness, but service mix and customer selection may confound these associations. Any bundle or contract intervention should be tested among comparable customers.

3.4 Cross-factor synthesis and weak signals

To compare grouped and categorical variables on a common descriptive basis, the analysis calculated the difference between the highest and lowest observed churn rates within each variable. Service calls, contract type and complaints produced the largest spreads, followed by satisfaction and late payments. Demographic variables remained below 0.3 percentage points.

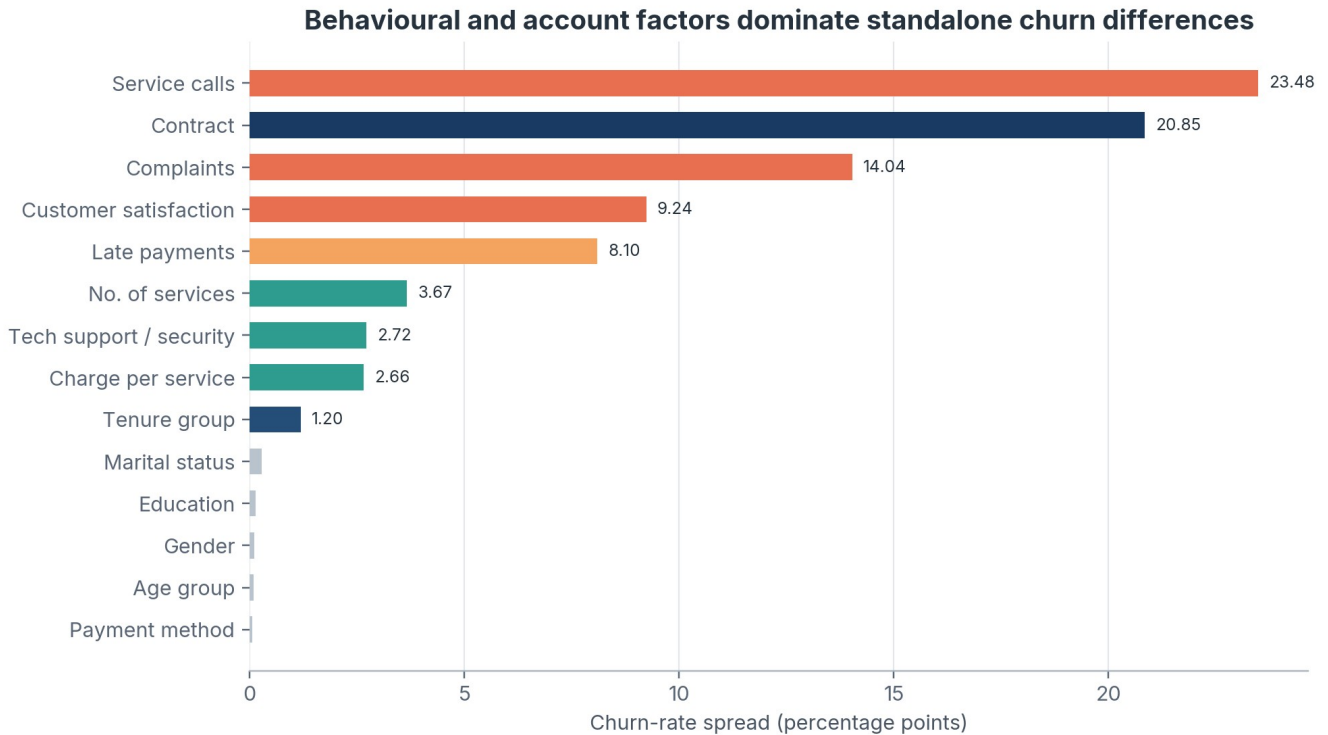


Figure 6. Descriptive churn-rate spreads. Extreme categories and small samples can affect rankings, so this is not equivalent to formal model importance.

Variables with little standalone value

Variable	Key result
Age group	9.90%-9.99% churn
Gender	9.82%-9.93% churn
Education	9.86%-10.00% churn
Marital status	9.70%-9.98% churn
Annual income	\$58,488 retained vs \$58,545 churned mean
Payment method	9.88%-9.95% churn
Credit score	678.63 retained vs 678.52 churned mean
Days since interaction	44.49 retained vs 44.55 churned mean
Average usage	38.64 GB retained vs 37.75 GB churned mean

EDA conclusion

Observable customer experience and account information carried substantially more standalone signal than demographics. The strongest candidates for joint modelling were contract type, complaints, service calls, satisfaction, support/security services, service count, late payments and charge per service.

4. Feature Engineering and Modelling Approach

The modelling stage tested whether the strongest univariate EDA patterns remained useful when variables were considered jointly. The sign-up date was excluded because customer lifecycle information was already represented by tenure. Monthly charge per service was recreated as the main engineered feature.

Feature preparation

Stage	Implementation
Starting cleaned data	1,000,000 rows x 31 variables
Engineered feature	monthly_charge_per_service = monthly charges / number of services
Categorical encoding	One-hot encoding with the first category removed
Final predictor matrix	39 encoded predictors
Training set	800,000 customers
Test set	200,000 customers
Class balance	9.92% churn in both sets through stratified sampling

Models and imbalance handling

Model	Role	Key configuration
Logistic Regression	Interpretable linear baseline	Standardised inputs; class_weight="balanced"
Decision Tree	Non-linear rule-based model	max_depth=8; class_weight="balanced"
Random Forest	Ensemble of 200 trees	max_depth=12; class_weight="balanced"
Gradient Boosting	Sequential boosted trees	100,000-row training subsample; balanced sample weights
XGBoost	Regularised boosted-tree model	200 estimators; scale_pos_weight=9.08; row and column subsampling

Evaluation principle

Accuracy was not treated as the primary success metric because a naive model predicting every customer as retained would be about 90% accurate while identifying no churners. Recall, precision, F1-score and ROC-AUC were therefore compared, with recall prioritised because missing an actual churner may be more costly than an unnecessary outreach.

5. Model Performance and Selection

All five models performed within a narrow range. XGBoost produced the highest ROC-AUC at 0.6850, but Logistic Regression was only 0.0001 lower. Gradient Boosting achieved the highest recall at 65.05%. The limited spread across linear, single-tree, bagged and boosted approaches suggests that the available features set the main performance ceiling, not model complexity alone.

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
XGBoost	62.76%	15.85%	63.91%	25.40%	0.6850
Logistic Regression	62.76%	15.86%	63.95%	25.42%	0.6849
Gradient Boosting	61.30%	15.49%	65.05%	25.02%	0.6808
Random Forest	64.18%	15.90%	60.86%	25.21%	0.6776
Decision Tree	60.67%	15.14%	64.32%	24.51%	0.6704



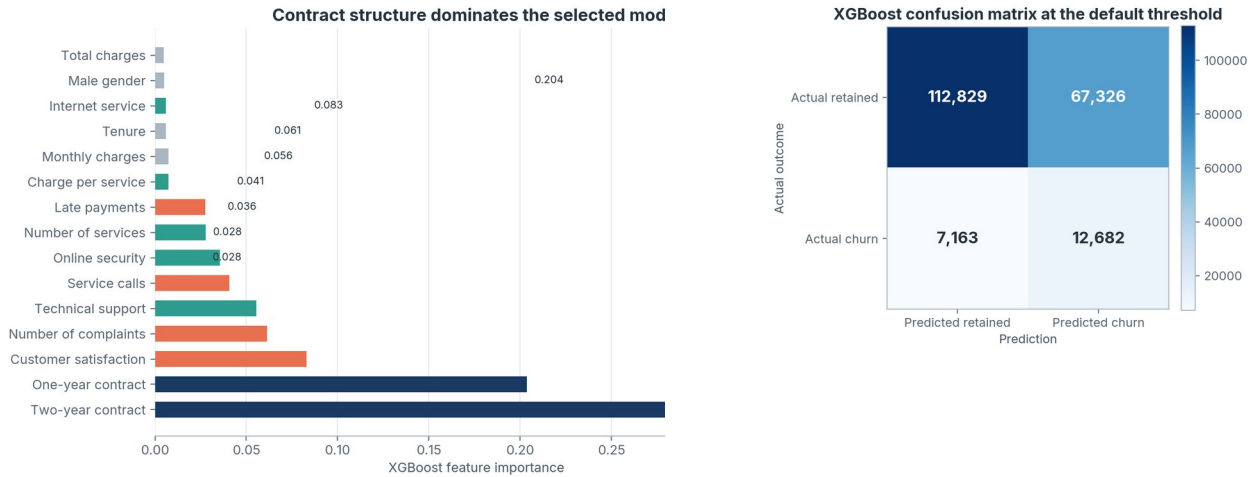
Figure 7. Test-set model comparison. Recall improved through class balancing, but precision remained around 15%-16%.

Model selection decision

XGBoost was selected for feature-importance interpretation because it had the highest ROC-AUC, not because it decisively outperformed the alternatives. Logistic Regression remained a competitive and more interpretable baseline. Cross-validation would be needed before claiming a statistically meaningful difference.

6. Model Interpretation

XGBoost feature importance largely confirmed the exploratory findings. The one-year and two-year contract indicators dominated the model, followed by customer satisfaction, complaints, technical support, service calls, online security, service count and late payments. Demographic variables contributed almost no importance.



Figures 8-9. XGBoost feature importance and confusion matrix at the default threshold. Importance is global and does not indicate causation.

The engineered charge-per-service feature ranked tenth, slightly above raw monthly charges and tenure. This suggests potential incremental value, but the result should be confirmed through an ablation comparison because correlated component variables can split or dilute tree-based importance.

Operational meaning

The model correctly identified 12,682 of 19,845 churners, but also flagged 67,326 retained customers. At the default threshold, it is better suited to ranking customers by risk than automatically assigning expensive retention interventions. Thresholds should differ for low-cost and high-cost actions.

7. Integrated Business Recommendations

1

Prioritise contract-based retention

Evidence: Month-to-month churn was 26.5%, and contract indicators ranked first and second in XGBoost.

Action: Test loyalty pricing, onboarding and voluntary movement to longer commitments while controlling for tenure and service mix.

2

Build a service-friction early warning layer

Evidence: Low satisfaction, complaint volume and repeated service calls were strong in both EDA and modelling.

Action: Combine these signals into a ranked alert; use service recovery outreach before offering discounts.

3

Test support and security bundles

Evidence: Technical support and online security were the strongest individual service signals and ranked in the model top 10.

Action: Test support-oriented bundles among comparable at-risk customers rather than assuming the association is causal.

4

Use late payments as an early warning signal

Evidence: Churn rose from 9.06% with no late payments to 17.16% with three or more; late payments ranked ninth.

Action: Include payment history alongside service-experience indicators, not as a stand-alone credit judgement.

5

Avoid demographic-first targeting

Evidence: Age, gender, education, marital status and income showed negligible EDA separation and model importance.

Action: Direct retention resources toward observable customer experience and account behaviour.

6

Match intervention cost to model threshold

Evidence: XGBoost recall was 63.9%, but precision was 15.9% at the default threshold.

Action: Use lower thresholds for cheap emails and higher thresholds for costly calls, discounts or account-manager actions.

8. Limitations and Future Improvements

Key limitations

Limitation	Implication
Synthetic data	Relationships may not generalise to a real telecommunications customer base.
Univariate EDA	Observed differences may reflect confounding among contract, tenure, service count and service mix.
Association rather than causation	Feature importance and churn-rate differences do not establish that changing a feature will reduce churn.
Modest discrimination	All models converged around ROC-AUC 0.67-0.69, indicating limited signal in the available features.
Low precision at default threshold	Many retained customers were flagged, preventing direct use for expensive interventions.
Random rather than temporal validation	The model was not tested on later customers than those used for training.
Gradient Boosting subsample	One model used 100,000 training records for computational efficiency, limiting direct comparability.
Importance limitations	Correlated variables can divide feature importance, and global rankings do not explain individual predictions.

Priority next steps

- Tune thresholds using the estimated cost of missed churners, unnecessary outreach and customer lifetime value.
- Add precision-recall AUC and precision-recall curves, which are especially informative under 9.92% churn prevalence.
- Use cross-validation and a separate validation set for model and threshold selection before final test evaluation.
- Build a reproducible scikit-learn Pipeline and ColumnTransformer with handle_unknown="ignore" for deployment-safe preprocessing.
- Apply SHAP analysis to explain individual customer risk and support targeted interventions.
- Test interaction features such as contract x complaints and satisfaction x service calls.
- Validate performance temporally and on real or independent datasets.
- Evaluate interventions through controlled experiments rather than relying on observational associations.

Recommended technical direction

The highest-value improvement is not simply a more complex model. It is a better decision framework: stronger behavioural data, leakage-safe validation, calibrated probabilities, threshold optimisation and cost-sensitive evaluation.

9. Conclusion

This project completed the full customer-churn workflow across data preparation, exploratory analysis and predictive modelling. One million customer records were cleaned and validated without discarding observations. The analysis consistently found that contract structure, customer satisfaction, complaint history, service-call frequency, support-oriented services, service adoption and late payments carried more useful churn information than demographic characteristics.

The modelling results reinforced the EDA but also established an important constraint: five materially different algorithms produced almost identical discrimination. XGBoost achieved the highest ROC-AUC at 0.6850, yet Logistic Regression was effectively tied. The main limitation is therefore the predictive information available in the features rather than the absence of a sufficiently advanced algorithm.

For business use, the strongest strategy is to rank customers using observable account and service-friction signals, then match the intervention to the risk score and economic cost. Month-to-month customers with low satisfaction, repeated complaints, frequent service calls or late payments should receive the earliest attention. Demographic-first campaigns should be deprioritised.

Final project outcome

The project demonstrates a complete and honest data-science process: strong data quality controls, business-led exploration, appropriate class-imbalance handling, transparent model comparison, cautious interpretation and actionable retention recommendations. The model is not yet deployment-ready, but it provides a credible foundation for cost-sensitive risk ranking and future experimentation.

Portfolio links

[GitHub profile](#) | [Customer churn repository](#) | [LinkedIn](#)

Appendix A. Technical Summary

A1. Strongest exploratory patterns

Factor	Lower-risk group	Higher-risk group	Spread
Contract	Two-year: 5.65%	Month-to-month: 26.51%	20.85 pp
Service calls	0 calls: 7.71%	10 calls: 31.19%	23.48 pp
Complaints	0: 8.10%	4+: 22.14%	14.04 pp
Satisfaction	Score 9: 7.48%	Score 1: 16.72%	9.24 pp
Late payments	0: 9.06%	3+: 17.16%	8.10 pp
Number of services	6: 7.37%	1: 11.04%	3.67 pp
Charge per service	Lowest decile: 8.61%	Highest decile: 11.27%	2.66 pp

A2. Top XGBoost features

Rank	Feature	Importance
1	Two-year contract	0.335586
2	One-year contract	0.203735
3	Customer satisfaction	0.083098
4	Number of complaints	0.061362
5	Technical support	0.055642
6	Service calls	0.040780
7	Online security	0.035672
8	Number of services	0.027714
9	Late payments	0.027567
10	Charge per service	0.007465

A3. Reproducibility notes

- Random seed: 42 for train/test split and model configurations where specified.
- Stratified split: 80% training and 20% testing, preserving 9.92% churn.
- Logistic Regression inputs were standardised; tree-based models used unscaled predictors.
- Categorical variables were encoded with pandas one-hot encoding and drop_first=True.
- XGBoost class weight ratio: 9.08 negative customers per positive customer.
- Gradient Boosting was trained on a reproducible 100,000-row subsample.